

Problem Sheet for CS402/MA492, Semester 2, 2018/2019

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1. The letter N was found to be the most frequent letter in a large ciphertext produced using a shift cipher over the alphabet $ABC \dots Z$. Decrypt the following portion of the ciphertext:

CQRBRBJFNJTLRYQNA

2. A Vigenère cipher over the 27-letter alphabet $ABC \dots Z$ was used to produce the ciphertext

SDCOZKRPFNSDKMDELEYSRILGSRRESEMPRRQQWQTECDZSN₁ELSRMFW₁GRUD
IVDKOLQSOW₁YIRWDIXIREEDLMROEEHD₁JRILLSHRMJDULSNOIRXZAED₁DUO
FPVDDAWIRXZODIRTWOPWDVAZGWDSPDDEPBQDKLWYPTDIRHMTGBRF₁VRER
WTMCI

The corresponding plaintext begins A₁LONG₁LONG₁TIME₁AGO

Find the key that was used and decrypt the message.

3. Identify $\mathbb{Z}_{26} = \{0, \dots, 25\} = \{A, \dots, Z\}$. An affine cipher with unknown key $(a, b) \in \mathbb{Z}_{26}^\times \times \mathbb{Z}_{26}$ over $\mathbb{Z}_{26}$ encrypts $E \rightarrow R$ and $S \rightarrow H$. Determine (a, b) .
4. Determine the size of the key space for an affine cipher over $\mathbb{Z}_n$ in each of the following two cases:
 - (a) n is a prime.
 - (b) $n = 2^k$ for some $k \geq 0$.

5. Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \in M_{2 \times 2}(\mathbb{Z}_n)$. The **adjugate** of A is the matrix

$$\text{adj}(A) := \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} \in M_{2 \times 2}(\mathbb{Z}_n).$$

- (a) Show that $A \cdot \text{adj}(A) = \text{adj}(A) \cdot A = \det(A) \cdot \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$.
 - (b) Show that if $\bar{d} \in \mathbb{Z}_n$ satisfies $\det(A)\bar{d} = 1$, then $\bar{d}\text{adj}(A)$ is an inverse of A over $\mathbb{Z}_n$.
 - (c) Show that if $\det(A) \notin \mathbb{Z}_n^\times$, then A is *not* invertible over $\mathbb{Z}_n$.
6. Identify $\mathbb{Z}_{26} = \{0, \dots, 25\} = \{A, \dots, Z\}$. A 2-dimensional Hill cipher over the 26-letter alphabet $ABC \dots Z$ was used to produce the ciphertext **ESDCWNMH** using the encryption function

$$\mathbb{Z}_{26}^2 \rightarrow \mathbb{Z}_{26}^2, \quad x \mapsto Ax,$$

where

$$A = \begin{bmatrix} 2 & 3 \\ 7 & 5 \end{bmatrix}.$$

Compute an inverse of A over $\mathbb{Z}_{26}$ and use it to find the plaintext.

7. Identify $\mathbb{Z}_{26} = \{0, \dots, 25\} = \{A, \dots, Z\}$. Suppose that the ciphertext digraphs DE and AJ correspond to the plaintext digraphs TH and HE, respectively, under a 2-dimensional Hill cipher over $\mathbb{Z}_{26}$. Recover the matrix used for encryption.
8. Consider a cryptosystem with sets $\mathcal{P}$, $\mathcal{C}$, and $\mathcal{K}$ of plaintexts, ciphertexts, and keys, respectively. As in Lecture 8, let X , Y , and K be associated random variables, where we choose plaintexts and keys independently at random. As usual, let $e_k: \mathcal{P} \rightarrow \mathcal{C}$ and $d_k: \mathcal{C} \rightarrow \mathcal{P}$ be the encryption and decryption function associated with the key $k \in \mathcal{K}$, respectively. Write $e_k[\mathcal{P}] := \{e_k(x) : x \in \mathcal{P}\}$. Show that for each $y \in \mathcal{C}$,

$$\text{Prob}(Y = y) = \sum_{\substack{k \in \mathcal{K} \text{ with} \\ y \in e_k[\mathcal{P}]}} \text{Prob}(K = k) \cdot \text{Prob}(X = d_k(y)).$$

9. Let $f: \mathbb{Z}_n^L \rightarrow \mathbb{Z}_n$ be an arbitrary function and let $s_0, \dots, s_{L-1} \in \mathbb{Z}_n$ be arbitrary. For $j \geq L$, we recursively define

$$s_j = f(s_{j-1}, \dots, s_{j-L}).$$

Show that the sequence $(s_0, s_1, \dots)$ is **ultimately periodic**: there are integers $C \geq 0$ and $N \geq 1$ such that

$$s_j = s_{j+N}$$

for all $j \geq C$.

10. A linear feedback shift register L of length 2 is used to encrypt messages using a stream cipher. It is known that when the cipher is used to encrypt the plaintext string $(x_0, x_1, x_2, x_3) = (0, 0, 1, 1)$, then the corresponding ciphertext string is $(y_0, y_1, y_2, y_3) = (1, 1, 1, 0)$. Find the connection polynomial of L .
11. Consider a linear feedback shift register L of length 5 with connection polynomial
- $$C(X) = 1 + X + X^5.$$
- (a) Given the initial state $(s_4, s_3, s_2, s_1, s_0) = (1, 1, 1, 0, 1)$, determine the least period of the resulting infinite sequence $(s_0, s_1, \dots)$.
- (b) Given the initial state $(s_4, s_3, s_2, s_1, s_0) = (0, 1, 1, 1, 0)$, determine the least period of the resulting infinite sequence $(s_0, s_1, \dots)$.
- (c) Write $C(X)$ as a product of irreducible polynomials over $\mathbb{Z}_2$. Which of these irreducible polynomials are primitive?

- (d) Find the maximal least period of a sequence associated with L as above over all possible initial states.

12. Let $n \geq 1$. Recall that

$$\mathbb{Z}_n^\times = \{a \in \mathbb{Z}_n : \gcd(a, n) = 1\}.$$

An integer $g \in \mathbb{Z}$ is called a **primitive root modulo n** if for each $a \in \mathbb{Z}_n^\times$, there exists an integer $k \geq 0$ with $g^k \equiv a \pmod{n}$.

- (a) Find a primitive root modulo n for $n = 2, 3, 4, 5, 6, 7, 9, 10, 11$.
(b) Show that there does not exist a primitive root modulo n for $n = 8, 12$.

Note. Primitive roots play a crucial role in the **Diffie-Hellman key exchange** protocol that we will discuss later.

13. Let $(n, e) = (65, 29)$ be a public RSA key.

- (a) Find the ciphertext corresponding to the message $m = 8 \in \mathbb{Z}_{65}$.
(b) Find the ciphertext corresponding to the message $m = 12 \in \mathbb{Z}_{65}$.
(c) Determine the private key (n, d) associated with (n, e) .

14. Let p be a prime.

- (a) Show that $\varphi(p^a) = p^a - p^{a-1}$ for each $a \geq 1$.
(b) Show that $x^p \equiv x \pmod{p}$ for each $x \in \mathbb{Z}$.

15. (a) Compute $\varphi(5424)$.

- (b) Show that if $n \geq 2$ is an even integer, then $\varphi(n) \leq \frac{n}{2}$.

16. Let $B \geq 1$ and $k \geq 0$ be integers. For $j = 0, \dots, k$, let a_j be an integer with $0 \leq a_j < B$. Let

$$A = a_0 + a_1B + \dots + a_kB^k.$$

Show that $A \leq B^{k+1} - 1$ with equality if and only if $a_0 = \dots = a_k = B - 1$.

17. Recall that a **reduced residue system modulo n** is a set $R \subset \mathbb{Z}$ with the following properties:

- $\gcd(r, n) = 1$ for all $r \in R$.
- For $a \in \mathbb{Z}$ with $\gcd(a, n) = 1$, there exists a unique $r \in R$ with $a \equiv r \pmod{n}$.

Let R be a reduced residue system modulo n and let $a \in \mathbb{Z}$ with $\gcd(a, n) = 1$. Show that $\{ar : r \in R\}$ is also a reduced residue system modulo n .

18. Let $n \geq 1$ be a pseudoprime to the base $a \in \mathbb{Z}$. Show that $\gcd(a, n) = 1$.

19. Let $g \in \mathbb{Z}_n$. Show that $\mathbb{Z}_n = \langle g \rangle_{\mathbb{Z}_n}$ (i.e. g is a generator of the additive(!) group $\mathbb{Z}_n$) if and only if $\gcd(g, n) = 1$.

20. Find the orders of the different elements of $\mathbb{Z}_7^\times$. Which of these elements are generators of the group?
21. Alice and Bob use the cyclic group $\mathbb{Z}_{47}^\times$ and its generator $g = 5$ to perform a Diffie-Hellman key exchange. Alice chooses $a = 4$ and Bob chooses $b = 2$. Determine their shared key.
22. Determine the set of points of the elliptic curve $y^2 = x^3 - x$ over $\mathbb{Z}_5$.